



Giant Ocean Waves More Common Than Thought

AP Associated Press

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Last year's Hurricane Ivan generated an ocean wave that towered higher than 90 feet at one point, says a study that also suggests such giants may be more common than once thought.

Research indicates these are not "rogue waves but actually fairly common during hurricanes," said David Wang of the Naval Research Laboratory at Stennis Space Center, Miss.

The giant wave was detected 75 miles south of Gulfport, Miss., by instruments on the ocean floor that measure the pressure of water above them. Using those readings, scientists can calculate the height of waves from trough to crest.

Last Sept. 15, as Hurricane Ivan passed through the area, the instruments measured 146 large waves, including 24 higher than 50 feet and one at 91 feet, Wang and his colleagues report in Friday's issue of the journal Science.

The giant wave did not reach land. Unlike a tsunami, which reaches down to the sea floor, this was a wind wave, generated on the ocean surface by the powerful forces of the storm.

Because shipping tends to try to avoid hurricanes, many large waves are unseen by humans, let alone measured.

Scientists at the National Oceanic and Atmospheric Administration have a different way of calculating wave heights, using buoys at sea.

Hendrik Tolman, an ocean wave expert at the NOAA's Climate Prediction Center, said a wave such as the giant one measured during Ivan is within expected limits.

Ocean researchers generally focus on "significant wave height," which is the average of the highest one-third of waves, he said. Within that average, there can be a much larger waves.

The highest significant wave height in Ivan was 52 feet as calculated by the NOAA buoys and 58 feet as calculated by Wang's group.

In a short-lived storm such as Ivan, a maximum wave of twice the significant height can occur, said Tolman, who was not part of Wang's research group.

Wang, however, said Ivan's towering wave exceeds those measured in other infamous storms.

"In 1969, Hurricane Camille produced a 44-foot wave by an oil rig near the storm's center," he said. "Only two other buoy reports exceed the 52-foot mark set by Ivan, both of which occurred in the North Pacific where winter storms are larger than hurricanes," Wang said.

With forecasters expecting continued high hurricane activity in the next few years, this report should be a good starting point to increase wave-height research, Wang said.

On Tuesday, meteorologists at the National Weather Service increased their storm forecast for this year. There have already been eight named storms and they said there could be as many as 11 to 14 more tropical storms, including seven to nine more hurricanes, by the end of November.

Also, on Monday, Kerry Emmanuel of the Massachusetts Institute of Technology published a report indicating that global warming is making hurricanes stronger.

Wang's research was funded by the Office of Naval Research and the Interior Department's Minerals Management Service, which oversees offshore oil drilling.

On the Net:

Science: <http://www.sciencemag.org>

National Oceanic and Atmospheric Administration: <http://www.noaa.gov>

Naval Research Laboratory: <http://www.nrlssc.navy.mil>

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